Week 4 Quiz

Graded

Student

[redacted]

Total Points

9 / 12 pts

Question 1

1a

1 / 3 pts

+ 3 pts 3 pts: Meets Expectations -
The solutions provide a sufficient explanation of the methods used to solve the problem. (e.g. the solutions apply theorems appropriately, and justify any intermediary results).
There are no arithmetic errors nor minor gaps in logic.

+ 2.999 pts Meets Expectations - There are only minor arithmetic errors or minor flaws in logic.

+ 2 pts 2pts: Approaches Expectations -
The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

✔ **+ 1 pt** 1 pt: Revision Needed -
The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts 0pts Not Assessable -
The solutions do not provide any logical explanations or justifications of the methods used to solve the problem.
There may be several major errors or gaps in logic.

Good thought! We want u to be what we need to remove from the integral so ln(x). Additionally, we integrate ln(x) to get v, but we need an integration by parts to even do that! Remember LIATE as your choice for u:)

Question 2

1b

3 / 3 pts

✔ **+ 3 pts** 3 pts: Meets Expectations -
The solutions provide a sufficient explanation of the methods used to solve the problem. (e.g. the solutions apply theorems appropriately, and justify any intermediary results).
There are no arithmetic errors nor minor gaps in logic.

+ 2.999 pts Meets Expectations - There are only minor arithmetic errors or minor flaws in logic.

+ 2 pts 2pts: Approaches Expectations -
The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

+ 1 pt 1 pt: Revision Needed -
The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts 0pts Not Assessable -
The solutions do not provide any logical explanations or justifications of the methods used to solve the problem.
There may be several major errors or gaps in logic.

For your work in part a, your integration by parts looks good!

Question 3

2a

3 / 3 pts

✔ **+ 3 pts** 3 pts: Meets Expectations -
The solutions provide a sufficient explanation of the methods used to solve the problem. (e.g. the solutions apply theorems appropriately, and justify any intermediary results).
There are no arithmetic errors nor minor gaps in logic.

+ 2.999 pts Meets Expectations - There are only minor arithmetic errors or minor flaws in logic.

+ 2 pts 2pts: Approaches Expectations -
The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

+ 1 pt 1 pt: Revision Needed -
The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts 0pts Not Assessable -
The solutions do not provide any logical explanations or justifications of the methods used to solve the problem.
There may be several major errors or gaps in logic.

Question 4

2b

2 / 3 pts

+ 3 pts 3 pts: Meets Expectations -
The solutions provide a sufficient explanation of the methods used to solve the problem. (e.g. the solutions apply theorems appropriately, and justify any intermediary results).
There are no arithmetic errors nor minor gaps in logic.

+ 2.999 pts Meets Expectations - There are only minor arithmetic errors or minor flaws in logic.

✔ **+ 2 pts** 2pts: Approaches Expectations -
The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

+ 1 pt 1 pt: Revision Needed -
The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts 0pts Not Assessable -
The solutions do not provide any logical explanations or justifications of the methods used to solve the problem.
There may be several major errors or gaps in logic.

Good setup but you are missing bounds for your integral!

Math 102, 10am **Week 3 Quiz** **Feb 3, 2025**

Name: [redacted] **Student ID:** [redacted]

Show all work and clearly denote your final answer. No credit will be given if there is no work shown. No calculators or outside help is permitted on this quiz.

1. Consider the following integral:

$$\int x^2 \ln(x)\, dx$$

(a) Set up integration by parts by choosing an appropriate u and dv. Compute du and v.

$U = x^2$ $dv = \ln(x)\, dx$

$du = 2x\, dx$ $V = \frac{1}{x}$

(b) Compute $\int x^2 \ln(x)\, dx$ using your answer to part (a).

$\int x^2 \ln(x)\, dx = UV - \int v\, dU$

$= x - \int \frac{1}{x} \cdot 2x\, dx = \int 2\, dx$

$= 2x$

$x - 2x$

$\boxed{\int x^2 dx = -x}$

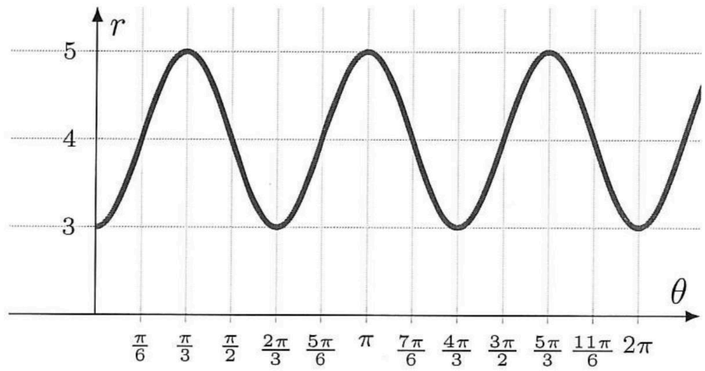
[redacted] [redacted]

Math 102, 10am Week 3 Quiz Feb 3, 2025

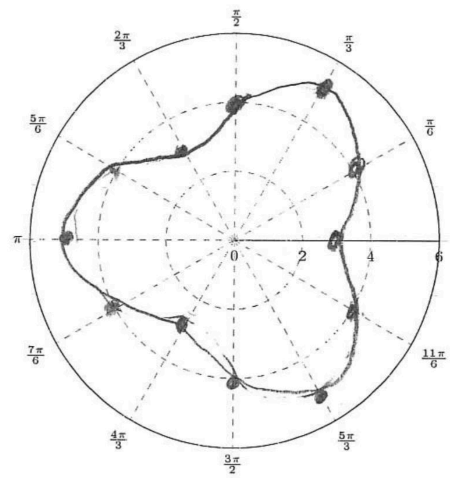
2. Consider the polar curve

$$r = 4 - \cos(3\theta)$$

with graph in the Cartesian plane below



(a) Sketch the polar graph of $r = 4 - \cos(3\theta)$.



(b) Set up, **but do not evaluate**, an integral that computes the area bounded by $r = 4 - \cos(3\theta)$.

$A = \frac{1}{2}\int r^2 \, d\theta$

$\boxed{A = \frac{1}{2}\int (4 - \cos(3\theta))^2 \, d\theta}$